

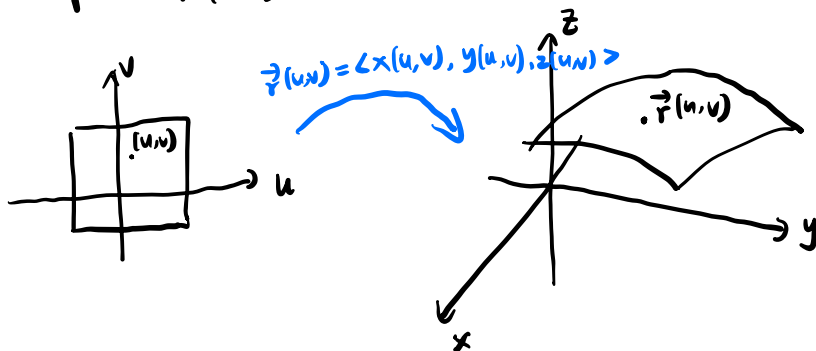
Lecture 20 Surface Integrals

Recap: $\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\nabla \times \vec{F}) \cdot \vec{k} dA$.

Surface parametrization.

Curve: $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$.

Surface: $\vec{r}(u,v) = x(u,v)\vec{i} + y(u,v)\vec{j} + z(u,v)\vec{k}$.



Example. The cylinder with radius r_0 and with z -axis as the central axis.

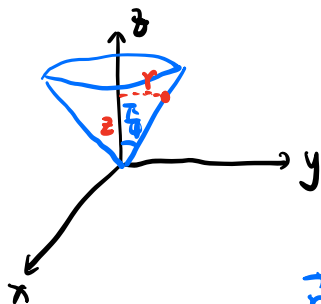
Sol. Cylindrical coordinates: $r = r_0$

$$\begin{cases} x = r_0 \cos \theta \\ y = r_0 \sin \theta \\ z = z \end{cases}$$

$$(r, \theta, z) \rightarrow (x, y, z)$$

$$\vec{r}(\theta, z) = (r_0 \cos \theta)\vec{i} + (r_0 \sin \theta)\vec{j} + z\vec{k} \quad 0 \leq \theta \leq 2\pi \quad -\infty < z < \infty$$

Example. A cone making $\frac{\pi}{4}$ angle with the z -axis.



Sol. Cylindrical coordinates: $z = r$.

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = r \end{cases}$$

$$\vec{r}(r, \theta) = (r \cos \theta)\vec{i} + (r \sin \theta)\vec{j} + r\vec{k} \quad 0 \leq r < \infty \quad 0 \leq \theta < 2\pi$$

Spherical coordinates (ρ, θ, ϕ) $\phi = \frac{\pi}{4}$

$$\begin{cases} x = \rho \sin \phi \cos \theta = \frac{\rho \cos \theta}{\sqrt{2}} \\ y = \rho \sin \phi \sin \theta = \frac{\rho \sin \theta}{\sqrt{2}} \\ z = \rho \cos \phi = \frac{\rho}{\sqrt{2}} \end{cases}$$

$$\vec{r}(\rho, \theta) = \left(\frac{\rho \cos \theta}{\sqrt{2}} \right) \vec{i} + \left(\frac{\rho \sin \theta}{\sqrt{2}} \right) \vec{j} + \left(\frac{\rho}{\sqrt{2}} \right) \vec{k} \quad 0 \leq \rho < \infty, \quad 0 \leq \theta \leq 2\pi.$$

Example. A sphere with radius 3

Sol. Spherical coordinates: $\rho = 3$

$$\vec{r}(\theta, \phi) = (3 \sin \phi \cos \theta) \vec{i} + (3 \sin \phi \sin \theta) \vec{j} + (3 \cos \phi) \vec{k}.$$

$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi.$

Example. A surface given by $x^2 + y - z^3 = 1$

Sol. $y = z^3 - x^2 + 1$

$$\vec{r}(x, z) = x \vec{i} + (z^3 - x^2 + 1) \vec{j} + z \vec{k}$$

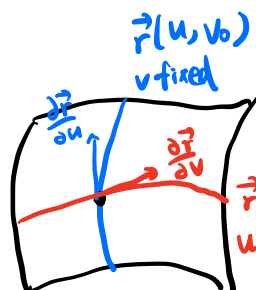
$$x = \pm \sqrt{z^3 - y + 1}$$

$$z = (x^2 + y - 1)^{1/3}$$

$$\vec{r}(x, y) = x \vec{i} + y \vec{j} + (x^2 + y - 1)^{1/3} \vec{k}.$$

Geometry of parametric surfaces

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$



$$\frac{\partial \vec{r}}{\partial u} = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right\rangle = \text{tangent vector to the blue curve}$$

$$\frac{\partial \vec{r}}{\partial v} = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right\rangle = \text{tangent vector to the red curve}$$

Both $\frac{\partial \vec{r}}{\partial u}$ and $\frac{\partial \vec{r}}{\partial v}$ are tangent to the surface

$\Rightarrow$ their cross product is normal to the surface.

A unit normal at each point is $\vec{n} = \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|}$

Surface integrals.

$$\iint_R f(x,y) \overset{dA}{dx dy}$$

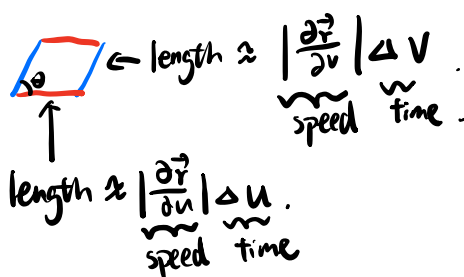
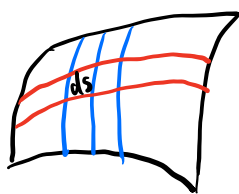
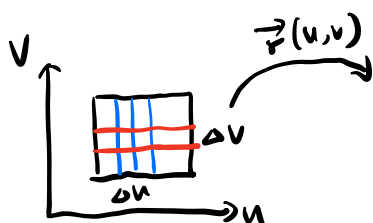
Def. (Surface Integrals) Given a surface S parametrized by $\vec{r}(u,v)$ with $a \leq u \leq b$ and $c \leq v \leq d$, and a continuous, scalar-valued function $f(x,y,z)$, the surface integral of f over the surface S is denoted by

$$\iint_S f dS = \int_{v=c}^{v=d} \int_{u=a}^{u=b} f(\vec{r}(u,v)) \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv$$

Notation: If S is a closed surface, $\oiint_S$

Surface element.

$$dS = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv$$



$$dS \approx \text{area of this parallelogram} = \left| \frac{\partial \vec{r}}{\partial u} \right| \Delta u \cdot \left| \frac{\partial \vec{r}}{\partial v} \right| \Delta v \cdot \sin \theta$$

$$= \left| \frac{\partial \vec{r}}{\partial u} \right| \left| \frac{\partial \vec{r}}{\partial v} \right| \sin \theta \Delta u \Delta v$$

$$= \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \Delta u \Delta v$$

$$\Delta u \cdot \Delta v \rightarrow 0 \quad dS = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv$$

$$\iint_S f \, ds = \begin{cases} \text{if } f=1, & \text{then this is the total surface area} \\ \text{if } f \text{ is the surface density,} & \text{then this is the total mass of the surface.} \end{cases}$$

Surface integrals on explicitly defined surfaces.

$$z = g(x, y)$$

$$\vec{r} = \langle x, y, g(x, y) \rangle$$

$$\frac{\partial \vec{r}}{\partial x} = \langle 1, 0, \frac{\partial z}{\partial x} \rangle \quad \frac{\partial \vec{r}}{\partial y} = \langle 0, 1, \frac{\partial z}{\partial y} \rangle$$

$$\frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & \frac{\partial z}{\partial x} \\ 0 & 1 & \frac{\partial z}{\partial y} \end{vmatrix} = \langle -\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1 \rangle.$$

$$\left| \frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right| = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$$

$$\iint_S f(x, y, z) \, ds = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dx \, dy.$$